

Math 289, Fall 2026

Homework 5

Due Wednesday, October 7, 2026, 3:00pm, on Gradescope

Problem 1. Compute the n th power of the $m \times m$ matrix

$$J_m(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & \lambda \end{pmatrix}, \quad \lambda \in \mathbb{C}.$$

Problem 2. Compute the determinant of the $n \times n$ matrix $A = (a_{ij})_{ij}$, where

$$a_{ij} = \begin{cases} (-1)^{|i-j|} & \text{if } i \neq j, \\ 2 & \text{if } i = j. \end{cases}$$